

Tutorial 5 (21 April 2026)

1. A streaming platform manages and sells subscriptions. During any given month, a **Basic** subscriber has a 0.3 chance of upgrading to **Standard** and a 0.2 chance of cancelling the subscription. Similarly, a **Standard** subscriber has a 0.4 chance of upgrading to **Premium** and a 0.2 chance of cancelling the subscription. Finally, a **Premium** subscriber has a 0.2 chance of cancelling the subscription during any month.

The platform may acquire new subscribers directly into the Basic, Standard, or Premium categories through targeted marketing campaigns, or may also deliberately reduce subscribers in any category through account clean-up initiatives. Assume that each marketing campaign unit brings in 15 subscribers on average.

How many campaign units aimed at each type of subscriber, Basic, Standard, or Premium, should the platform launch or reduce every month, in order to maintain 300 Basic subscribers, 525 Standard subscribers, and 675 Premium subscribers at all time?

2. Customers arrive at a parcel collection counter according to a Poisson process at an average rate of **24 customers per hour**.
 - (a) What is the probability that **exactly 5 customers** arrive during the next **10 minutes**?
 - (b) What is the probability that **at least 6 customers** arrive during the next **15 minutes**?
 - (c) What is the probability that **at most 3 customers** arrive during the next **8 minutes**?
 - (d) Determine the **mean** and **standard deviation** of the number of customers arriving during the next **90 minutes**.
 - (e) Determine the probability that the time between two consecutive customer arrivals is **less than 1 minute**.
 - (f) Determine the probability that the time between two consecutive customer arrivals is **between 2 and 4 minutes**.
 - (g) Determine the probability that the time between two consecutive customer arrivals is **more than 5 minutes**.
 - (h) Given that **no customer has arrived during the last 3 minutes**, what is the probability that the next arrival occurs **within the next 2 minutes**?
 - (i) A manager claims that because the average arrival rate is 24 customers per hour, exactly 2 customers must arrive every 5 minutes on average. Briefly explain why this statement is misleading.
3. Inter-arrival times at a self-service coffee kiosk are exponentially distributed with an average time of **4 min** between arrivals. The duration of time that customers spend using the kiosk is also exponentially distributed with a mean duration of **3 min**. There is a large amount of space for queue formation by people waiting to use the kiosk, and arrivals are not dissuaded at all by lengthy queues.
 - (a) Determine the expected fraction of time that the kiosk will be in use.
 - (b) Determine the expected number of people in the kiosk system, that is, those waiting to use the kiosk together with the person using the kiosk, when applicable.

- (c) Determine the expected number of people waiting in the queue before being able to use the kiosk.
 - (d) How long should a kiosk customer expect to spend in the kiosk system in total, that is, the time spent waiting for the kiosk to become available together with the subsequent service time itself?
 - (e) What is the probability that an arrival will not have to wait at all before the kiosk becomes available?
 - (f) What is the probability that an arrival will have to wait between **2 and 5 minutes** before the kiosk becomes available?
 - (g) What is the probability that an arrival will have to wait more than **6 min** in the queue before the kiosk becomes available?
 - (h) What is the probability that exactly three people will be in the kiosk system in total?
 - (i) What is the probability that at least three people will be in the kiosk system in total?
 - (j) What is the probability that at most three people will be in the kiosk system in total?
4. A campus print station has a **single high-speed printer** used by students. Students arrive according to a Poisson process at an average rate of **15 students per hour**. The time taken by a student to complete printing is exponentially distributed with a mean of **3 minutes**. There is sufficient space for students to queue, and no students leave because of a long queue.
- (a) Give the Kendall-Lee notation for this queuing system.
 - (b) Determine the arrival rate λ , service rate μ , and traffic intensity ρ .
 - (c) Determine the expected fraction of time that the printer is busy.
 - (d) Determine the expected number of students in the system, L .
 - (e) Determine the expected number of students waiting in the queue, L_q .
 - (f) Determine the expected total time that a student spends in the system, W .
 - (g) Determine the expected waiting time in the queue, W_q .
 - (h) What is the probability that an arriving student will not have to wait at all before using the printer?
 - (i) What is the probability that exactly **four** students are in the system?
 - (j) What is the probability that **at least four** students are in the system?
 - (k) On average, how many students per hour complete printing?
 - (l) Suppose the station manager can replace the current printer with a faster printer that reduces the mean service time to **2 minutes**, but this costs an additional **R18 per hour**. Each hour that a student spends in the system is valued at **R25 per hour** in lost time. Which option should be recommended, based on **total expected hourly cost**?